

Table 5: Ablation study of different reward components in the RDFR training stage. Improvements are computed relative to Qwen2.5-Instruct-STIT.

Method	MSE ↓		MAE ↓	
Qwen2.5-Instruct (Zero-shot)	—		—	
Qwen2.5-Instruct-STIT	0.403		0.428	
+ R_{len}	0.401	(-0.50%)	0.425	(-0.70%)
+ R_{fmt}	0.395	(-1.99%)	0.419	(-2.10%)
+ R_{acc}	0.367	(-8.93%)	0.398	(-7.01%)
+ $R_{len} + R_{fmt}$	0.388	(-3.72%)	0.412	(-3.74%)
RDTU (R_{total})	0.351	(-12.90%)	0.382	(-10.75%)